

Value-at-Risk & Expected Shortfall Calculation (2024)

For each coin, calculated:

- 1. 90% Value-at-Risk (VaR): 10th percentile of the daily return distribution.
- 2. Expected Shortfall (ES): Average return below the 90% VaR.

Coins	BTC	DOGE	ETH	XRP	UNI
90% VaR	0.030031945	0.052918396	0.037127872	0.036124158	0.052279245
Expected Shortfall	0.0456135	0.081968256	0.056468978	0.06399471	0.084880373
Risk Ranking	1	5	3	2	4

While calculating VaR and ES for each coin, it became clear that not all crypto currencies carry the same risk profile. For example, Bitcoin consistently showed the lowest downside risk, aligning with its reputation as a relatively more stable asset within the crypto space. On the other hand, meme coins like DOGE displayed higher risk and deeper expected losses. Ranking the coins based on their VaR highlights how even within crypto, selecting the right combination of assets can help mitigate potential losses—a crucial insight when designing any risk-managed crypto portfolio.

Result: [BTC is the least risky; DOGE is the riskiest based on VaR](#)

Risk vs. Reward

To evaluate the efficiency of different cryptocurrencies in terms of risk and reward, Risk vs. Reward scatter plot was created. The daily expected shortfall was used as a proxy for risk, while the average daily return from 2024 data served as the measure for reward. The plotted assets include Bitcoin (BTC), Dogecoin (DOGE), Ethereum (ETH), Ripple (XRP), and Uniswap (UNI), along with a risk-free asset for comparison.



From the scatter plot, we observe the following trends:

- DOGE exhibits the highest risk and reward, indicating its potential for high returns but also substantial risk. Notably, DOGE appears more efficient than UNI, as both have similar risk levels, but DOGE offers a higher return.
- XRP also demonstrates a favorable risk-reward trade-off, making it one of the more efficient assets in this sample.
- BTC maintains a balanced risk-reward profile, offering moderate returns while keeping risk lower than most other assets, making it relatively more efficient than ETH.
- ETH, despite its prominence, appears less efficient in this dataset, as it has a lower return relative to its risk when compared to BTC, DOGE, and XRP.
- UNI, while showing relatively high returns, appears less efficient than DOGE, as it carries nearly the same level of risk but yields lower returns.

- The risk-free asset remains at the lowest end of both axes, serving as a benchmark with minimal risk and a stable but low return.

Efficiency and the Efficient Frontier

In the risk-reward trade-off, efficiency is determined by how close an asset is to the efficient frontier, a theoretical boundary representing the optimal portfolios that offer the highest return for a given level of risk.

In this analysis, DOGE and XRP emerge as more efficient options, as they deliver relatively higher returns for their respective levels of risk. ETH appears to be less efficient compared to BTC, indicating that Bitcoin provides a better risk-adjusted return. Similarly, UNI seems less efficient than DOGE, as both exhibit nearly the same level of risk, but DOGE offers a noticeably higher return.

Conclusion

This suggests that investors seeking an optimal balance between risk and return may find DOGE and XRP more attractive choices, while ETH and UNI may require further evaluation based on risk tolerance and investment goals.

Observation: [XRP and DOGE offer higher returns relative to their risk.](#)

Portfolio Risk

Based on the understanding of achieving a portfolio point on the most efficient frontier of the risk-reward diagram, the overall risk of a portfolio can be minimized through diversification. If a combination of coins can be invested in that reduces the shortfall to a level below that of investing solely in BTC, the lowest-risk coin, diversification will be pursued. No leverage has been used for the exercise.

Utilizing the invested coin weights as our variable input, and resulting daily portfolio returns from 2024, the minimized expected shortfall is derived by performing the following with the Crypto Data:

1. Ran excel solver for what-if analysis
2. Set objective to minimize the shortfall
3. Change variable cells as invested Crypto weights
4. The constraints used were
 - a. A minimum weight of 0 for each coin
 - b. The total weight of all coin investments equals 1

The coins and weights solved for are as follows:

W_BTC	W_DOGE	W_ETH	W_XRP	W_UNI
0.9648665	0.0000000	0.0000000	0.0351335	0.0000000

The return, value-at-risk, and expected shortfall of this portfolio in January 2025 are as follows:

90% VAR	0.0295616
Expected Shortfall	0.0454626
BTC (lowest individual risk)	0.0456135
Reduced shortfall risk by diversifying	0.0001509
January 2025 Return:	0.0035482

Previously as solved for in question 1, when evaluating the shortfall of the coins individually, BTC resulted as least risky with a shortfall of 0.0456135. When solving for variable investments in a single portfolio, thus diversifying, the least risky portfolio is derived from investing a small portion into XRP.

The resulting shortfall from the diversified portfolio is 0.0454626 which compared to only investing in our least risky coin BTC, has reduced our overall portfolio risk by 0.0001509. The strategy is to diversify from solely investing in BTC, to building our portfolio with 0.9648665 in BTC and 0.0351335 in XRP will successfully minimize the portfolio risk and derive an optimal risky portfolio based on reducing shortfall to the lowest level.

Result: [BTC + XRP minimizes downside risk by 0.0001509.](#)

TRO Strategy & Trade-off

With the same 2024 data, we maximized the risk-reward trade-offs ratio (Fig. 1) using the above formula solving for variable coin weights to implement the tail-risk optimized strategy. We utilized the Solver in a bid to determine the optimal allocation of the five cryptocurrencies (BTC, DOGE, ETH, XRP, and UNI).

Fig. 1

$$\text{Risk-Reward Trade-Off Ratio} = \frac{\text{Average Daily Return} - \text{Average Risk-Free Rate}}{\text{Daily Expected Shortfall}}$$

A similar what-if analysis to question 3 was performed, with the objective now set to maximize the risk-reward trade-off ratio. The coins and weights solved for are as follows:

W_BTC	W_DOGE	W_ETH	W_XRP	W_UNI
0.0000000	0.2437762	0.0000000	0.7562238	0.0000000

Using the above weights, we arrived at the following statistics:

90% VAR	0.038301184
Expected Shortfall	0.064091614
Average Daily Return	0.004412101
Average Risk-Free Rate	0.000140881
Risk-Reward Trade-Off Ratio	0.066642425

The most effective portfolio in maximizing the risk-reward trade-off ratio contains only two currencies: XRP (0.7562238) and DOGE (0.2437762)

The optimal portfolio weights for the highest trade-off ratio allocates weights to DOGE and XRP, as these coins provide relatively higher returns in comparison to other coins, despite the downside risk that is present. The resulting allocation reflects a well-balanced approach where the return sufficiently compensates for the potential losses captured by the expected shortfall. Interestingly, BTC, ETH, and UNI receive zero weight in the optimal solution, indicating that under the given conditions, they do not contribute to improving the risk-reward balance.

This highlights the significance of the risk-reward trade-off, which involves evaluating not just how much risk is present in a portfolio but also how effectively that risk is compensated by incremental returns. Rather than aiming to minimize risk alone, the focus is on achieving the highest possible return relative to the level of downside risk taken. Striking this balance is crucial in constructing a portfolio that is both efficient and resilient

TRO Strategy & Trade-off on February 1, 2025

We recalculated the optimal portfolio weights on February 1, 2025. A similar what-if analysis to question 4 was performed, using data from February 2024 through January 2025. The coins and weights solved for are as follows:

W_BTC	W_DOGE	W_ETH	W_XRP	W_UNI
0.0000000	0.2259857	0.0000000	0.7740143	0.0000000

Using the above weights, we arrived at the following statistics:

90% VAR	0.037976714
Expected Shortfall	0.065075612
Average Daily Return	0.005804668
Average Risk-Free Rate	0.000138560
Risk-Reward Trade-Off Ratio	0.087069604
February 2025 Return	(0.012358794)

The rebalanced portfolio recorded a negative return of 0.0123588 in February 2025, reflecting poor performance despite the optimization process. This result highlights an important insight: even though historical optimization may suggest an efficient portfolio structure, short-term market conditions can deviate significantly, leading to unexpected returns.

The model distribution remained concentrated in XRP and DOGE, with XRP's weight increasing moderately from 0.7562238 to 0.7740143 and DOGE's weight decreasing from 0.2437762 to 0.2259857. This minor rebalancing reflects relatively stable risk-reward profiles between these two cryptocurrencies over the 12-month rolling window.

Equally Weighted Portfolio

To determine if an equally weighted portfolio (allocating 20% to each cryptocurrency) outperforms the optimized portfolio from Question 5, we compared their risk-reward trade-off ratios and overall returns in February 2025.

Equal-Weighted Portfolio Performance (February 2025):

Risk-Free Rate	0.000140881
Average Daily Return	0.003898743
90% Value-at-Risk (VaR)	0.034693004
Expected Shortfall	0.059353576
Risk-Reward Trade-off Ratio	0.063313162
February 2025 Return	(0.011003534)

Comparison with Optimized Portfolio (Q5):

- The equally weighted portfolio achieved a lower risk-reward trade-off ratio (0.06330 vs. 0.08706) by 0.023756442.
- The equally weighted portfolio's return of -1.10% was better than the optimized portfolio's return in February 2025.
- The optimized portfolio was heavily concentrated in DOGE and XRP, while the equally weighted portfolio diversified risk more evenly across assets.

Conclusion:

The equally weighted portfolio outperformed the optimized portfolio in February 2025, demonstrating that optimization based on past data may not always lead to superior future performance. This suggests that a simpler diversification strategy could be more resilient in volatile market conditions. Additionally, while the equally-weighted portfolio exhibited a lower risk-reward trade-off ratio 0.063313162 compared to the optimized portfolio's higher trade-off ratio 0.087069604 based on historical data, its actual short-term performance turned out to be relatively more stable resulting in lower negative returns.

AUM & Rebalancing Portfolio

At the beginning of January 2025, assuming the fund invested \$100 million in assets under management (AUM). After portfolio growth in January 2025, the fund's AUM increased to \$135,206,999. A rebalancing was conducted at the beginning of February 2025 to adjust the portfolio weights to align with the updated strategy.

Rebalancing Adjustments (February 1, 2025):

	DOGE	XRP
Initial Portfolio Weights (Q4, January 2025)	0.2438	0.7562
Rebalanced Portfolio Weights (Q5, February 2025)	0.2259	0.7740

Rebalancing Transactions:

- DOGE: Need to **reduce (sell)** by \$2,405,400.
- XRP: Need to **increase (buy)** by \$2,405,400.

Fund Performance Post-Rebalancing:

- Total Loss since January 1, 2025: **(\$7,180,815)**
- Loss Since Rebalancing: **(\$42,387,814)**
- Final AUM at February EOM: \$92,819,185

Conclusion:

The portfolio required a rebalancing transaction of \$2.4 million, shifting funds from DOGE to XRP. However, despite rebalancing, the portfolio experienced a significant drawdown in February, leading to a decline in AUM. This highlights the challenges of frequent rebalancing in a volatile crypto market, where historical trends may not always predict future performance effectively.

Minimum Annual Management Fee

To determine the minimum annual management fee required to cover rebalancing costs, we assume:

1. A 3 basis points transaction fee on every dollar traded.
2. A 5 basis points bid-ask spread was applied when we purchased coins.
3. Monthly rebalancing, with January’s rebalance used as the basis for recurring monthly costs.

Based on the weight changes from January 1, 2025 (Q4 portfolio) to January 31, 2025 (Q5 portfolio), we calculated the total rebalancing costs incurred from adjusting DOGE and XRP allocations:

Coins	Initial Weight	End Weight	Change in Weights	Transaction Cost	Bid-Ask Cost	Total Cost
BTC	0	0	0	0	0	0
DOGE	0.2437762	0.2259857	0.0177905	5.33715E-06	0	5.33715E-06
ETH	0	0	0	0	0	0
XRP	0.7562238	0.7740143	0.0177905	5.33715E-06	8.89525E-06	1.42324E-05
UNI	0	0	0	0	0	0
Total Portfolio cost						1.95696E-05
Annualized Cost / Management cost						0.000234835

January Month:

- Total Monthly Rebalancing Cost: 0.000019569
- AUM: \$135,206,999
- Annual Fees : \$2,646

Annual Cost:

- % Fee: ~0.0235% of initial AUM.

By calculating the total transaction and bid-ask costs based on the January rebalancing activity, it becomes clear that even seemingly small adjustments in portfolio weights can accumulate meaningful expenses over time, especially with monthly rebalancing. The resulting minimum management fee of approximately **0.0235% of AUM annually** reflects a practical and sustainable way to cover these operational costs without compromising the portfolio's risk-return objectives. This reinforces the importance of not only optimizing portfolio allocations but also strategically planning fee structures to safeguard long-term investor value.